Reproducing Table I of “Tracking down Quirks at the LHC”

Knapen, Lou, Papucci & Setford, arXiv:1708.02243 · short report, version 2 · 3 October 2026

We redid the paper's signal and background study using only what the paper and its detector reference [27] say. Nothing was tuned to match the paper. Each simulated sample is shown on its own. Every setting is labelled by where it comes from.

What changed from version 1

- The **paper-stated** acceptance is used as the main result: barrel only, which the paper says “effectively restricts the fiducial range to $|\eta| \lesssim 1.8$”. The finite barrel lengths of reference [27] are now a labelled sensitivity study, since the paper never says it applied them.
- The fiducial denominators therefore return to **25** (colored, 2000-event sample) and **94** (EW, 2000-event sample).
- SCT resolution along the beam is now shown for every treatment: 1.16 mm (the paper's “twice the intrinsic accuracy” applied to the 580 µm that [27] quotes; inferred), 0.58 mm (sensitivity) and no smearing (undocumented). None of them is stated outright by the paper.
- The 45 mm beam-spot sentence allows three readings; all three are shown (section 5).

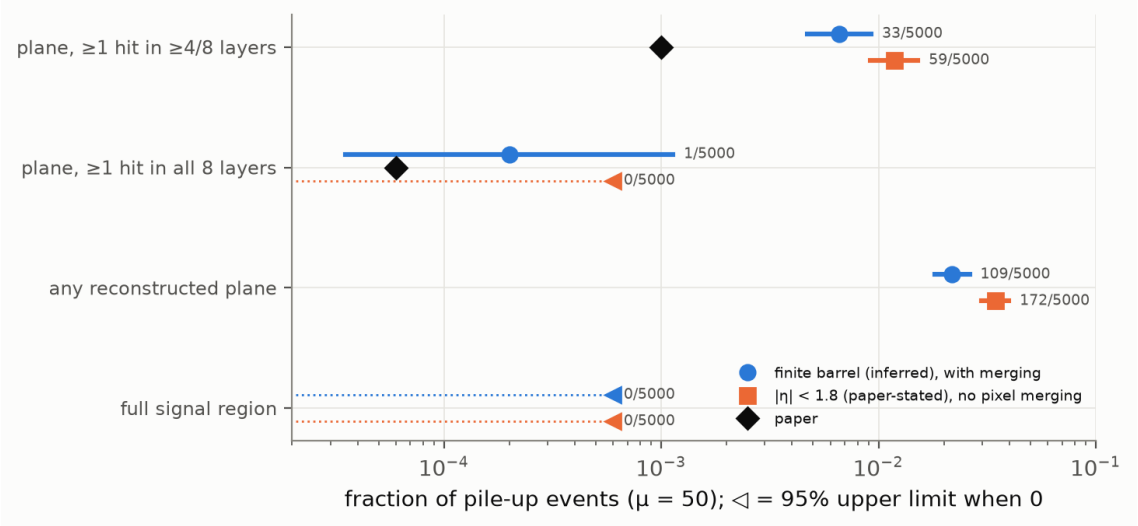
Exact p-value	What we call it
$p < 0.01$	**Strongly inconsistent**
$0.01 \le p < 0.05$	**Statistically inconsistent**
$0.05 \le p < 0.10$	**Borderline / mild tension**
$p \ge 0.10$	**Consistent**

How to read the tables. k/n = events passing / events tested. 95% CI = Wilson interval. With 0 events we quote the one-sided 95% upper limit (UL). p = exact binomial probability of our count if the paper's number were exactly right. The paper gives no counts or errors, so its own simulation uncertainty is ignored and borderline p-values are a little optimistic. Labels: **stated** = written in the paper or [27]; **inferred** = our reading of an unclear sentence; **undocumented** = no source.

1. Pile-up background ($\langle\mu\rangle = 50$, 5000 events per run)

Quantity	$\|\eta\| < 1.8$ run (paper-stated)	Finite-barrel run (sensitivity)	Paper	p ($\|\eta\|$ run)	Category
Plane with hits in ≥ 4 of 8 layers	$59/5000 = 1.2\times10^{-2}$	$33/5000 = 6.6\times10^{-3}$	$\sim1\times10^{-3}$	6.9×10^{-42}	**Strongly inconsistent**
Plane with hits in all 8 layers	0/5000 (UL 6.0×10^{-4})	$1/5000 = 2.0\times10^{-4}$	6×10^{-5}	1.00	**Consistent**
Any reconstructed plane	$172/5000 = 3.4\times10^{-2}$	$109/5000 = 2.2\times10^{-2}$	not given	—	**n/a**
Full signal region	0/5000 (UL 6.0×10^{-4})	0/5000 (UL 6.0×10^{-4})	“a few” in Fig. 5	—	**n/a**

The $|\eta| < 1.8$ run is the existing project run (SCT 34/1.16 mm, no pixel merging); the finite-barrel run includes merging. No new pile-up events were generated.



Pile-up rates with 95% intervals for both runs; diamonds = paper.

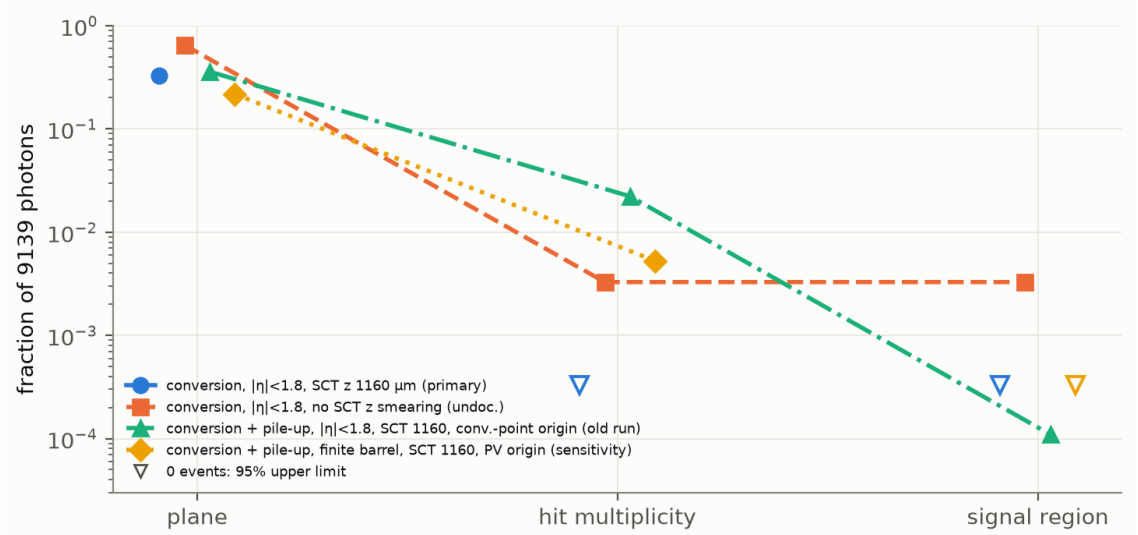
Reading

- No pile-up event reaches the signal region in either run (95% UL 6×10^{-4} per event), as the paper concludes.
- The all-8-layer rate agrees with the paper. The 4-of-8 rate is 6–12× higher. Likely reasons: our pile-up tracks are ideal helices with no energy loss, bremsstrahlung or endcaps; our pool has 20 000 minimum-bias events instead of 125 000; the paper does not define “a plane” there.

2. Converted-photon background (9139 Z(νν)γ+jet photons)

Setup (|η| < 1.8 unless stated)	Plane	Hit multiplicity	Signal region	Result
SCT z 1.16 mm (inferred)	3005/9139 = 32.9%	0/9139 (UL 3.3×10^{-4})	0/9139 (UL 3.3×10^{-4})	**No fake signal**
SCT z 0.58 mm (sensitivity)	5124/9139 = 56.1%	10/9139 = 0.11%	0/9139 (UL 3.3×10^{-4})	**No fake signal**
no SCT z smearing (undocumented)	5937/9139 = 65.0%	30/9139 = 0.33%	30/9139 = 0.33%	**30 fake signals**
+ pile-up, SCT 1.16 mm, fit from conversion point (old run)	3276/9139 = 35.8%	203/9139 = 2.2%	1/9139 (event 2555)	**1 fake signal**
+ pile-up, finite barrel, SCT 1.16 mm, fit from vertex	1990/9139 = 21.8%	48/9139 = 0.53%	0/9139 (UL 3.3×10^{-4})	**No fake signal**

No p-value: the paper gives no background rate to test against. Expected events at 300 fb^{-1} = σ × 1% × fraction × 300 fb^{-1}: with 0/9139 the 95% limit is < 0.21 events (our σ = 0.23 pb) or < 0.98 (paper σ ≈ 1 pb); 30/9139 gives ≈ 2.1 (our σ) to 9.8 (paper σ) events.



Fraction of photons surviving each cut. Open triangles: 95% upper limit when nothing survives.

Reading

- With the SCT smearing that [27] implies (1.16 or 0.58 mm), no conversion fakes a quirk, as the paper says.
- Without SCT z smearing, the setting that brings the signal closest to the paper (section 5), 30/9139 = 0.33% of conversions pass. That setting is not background-free.
- The old project overlay run had one passing event (2555), fitted from the conversion point; the paper fits from the primary vertex. In the finite-barrel run with the vertex fit, nothing passes. An overlay run with $|\eta| < 1.8$ and the vertex fit was not made (it needs a new 1.5 h pile-up pass).
- Our Z(νν)γ+jet cross-section is 4.3× below the paper's ~1 pb; generator settings are not given. The 1% conversion probability comes from the paper's own "~1 pb → ~10 fb".

3. Colored quirk signal (m_Q = 1800 GeV, Λ = 2 keV)

Efficiency	Sample	Ours	%	95% CI (%)	Paper	p	Category
Trigger	COL1000	101/419	24.1%	[20.3, 28.4]	0.24	0.95	**Consistent**
Trigger	COL2000	207/873	23.7%	[21.0, 26.6]	0.24	0.87	**Consistent**
Fiducial ($\|\eta\| < 1.8$)	COL1000	7/27	25.9%	[13.2, 44.7]	0.28	1.00	**Consistent**
Fiducial ($\|\eta\| < 1.8$)	COL2000	25/68	36.8%	[26.3, 48.6]	0.28	0.14	**Consistent**
Reco, SCT z 1.16 mm (inferred)	COL1000	1/7	14.3%	[2.6, 51.3]	0.35	0.43	**Consistent**
Reco, SCT z 1.16 mm (inferred)	COL2000	3/25	12.0%	[4.2, 30.0]	0.35	0.019	**Statistically inconsistent**
Reco, SCT z 0.58 mm (sensitivity)	COL1000	3/7	42.9%	[15.8, 75.0]	0.35	0.70	**Consistent**
Reco, SCT z 0.58 mm (sensitivity)	COL2000	5/25	20.0%	[8.9, 39.1]	0.35	0.14	**Consistent**
Reco, no SCT z smearing (undocumented)	COL1000	5/7	71.4%	[35.9, 91.8]	0.35	0.056	**Borderline / mild tension**
Reco, no SCT z smearing (undocumented)	COL2000	11/25	44.0%	[26.7, 62.9]	0.35	0.40	**Consistent**

Reconstruction denominators are the $|\eta| < 1.8$ fiducial events (7 and 25). Only the 1.16 mm row uses a value with a source; the other two rows show how much this one setting matters.

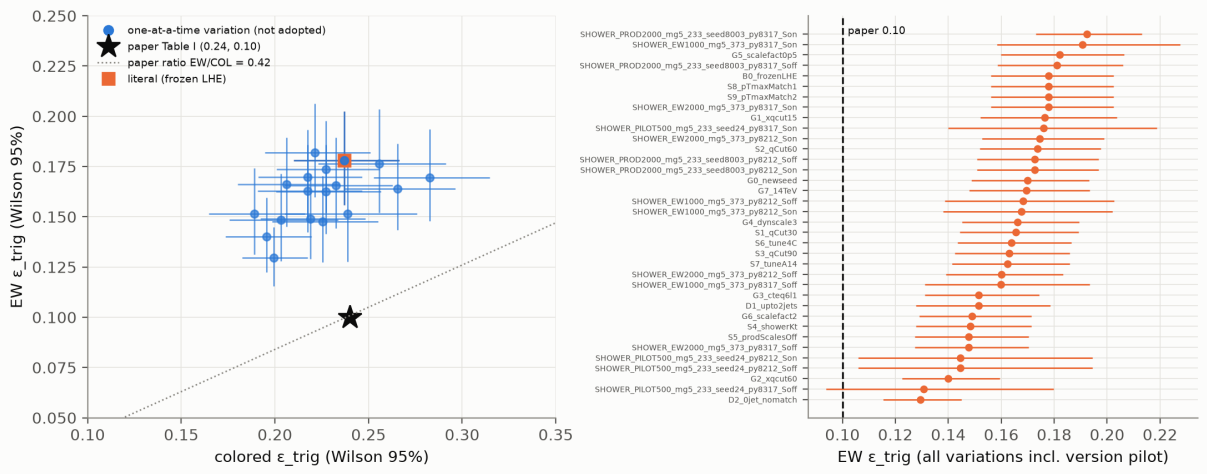
4. Electroweak quirk signal (m_Q = 800 GeV, Λ = 4 keV)

Efficiency	Sample	Ours	%	95% CI (%)	Paper	p	Category
Trigger	EW1000	89/511	17.4%	[14.4, 20.9]	0.10	2.8×10^{-7}	**Strongly inconsistent**
Trigger	EW2000	186/1045	17.8%	[15.6, 20.2]	0.10	1.8×10^{-14}	**Strongly inconsistent**
Fiducial ($\|\eta\| < 1.8$)	EW1000	40/89	44.9%	[35.0, 55.3]	0.28	8.2×10^{-4}	**Strongly inconsistent**
Fiducial ($\|\eta\| < 1.8$)	EW2000	94/186	50.5%	[43.4, 57.6]	0.28	1.3×10^{-10}	**Strongly inconsistent**
Reco, SCT z 1.16 mm (inferred)	EW1000	6/40	15.0%	[7.1, 29.1]	0.72	9.1×10^{-14}	**Strongly inconsistent**
Reco, SCT z 1.16 mm (inferred)	EW2000	21/94	22.3%	[15.1, 31.8]	0.72	2.4×10^{-23}	**Strongly inconsistent**
Reco, SCT z 0.58 mm (sensitivity)	EW1000	10/40	25.0%	[14.2, 40.2]	0.72	9.4×10^{-10}	**Strongly inconsistent**
Reco, SCT z 0.58 mm (sensitivity)	EW2000	29/94	30.9%	[22.4, 40.8]	0.72	1.5×10^{-16}	**Strongly inconsistent**
Reco, no SCT z smearing (undocumented)	EW1000	34/40	85.0%	[70.9, 92.9]	0.72	0.078	**Borderline / mild tension**
Reco, no SCT z smearing (undocumented)	EW2000	76/94	80.9%	[71.7, 87.5]	0.72	0.065	**Borderline / mild tension**

Reconstruction denominators: 40 and 94.

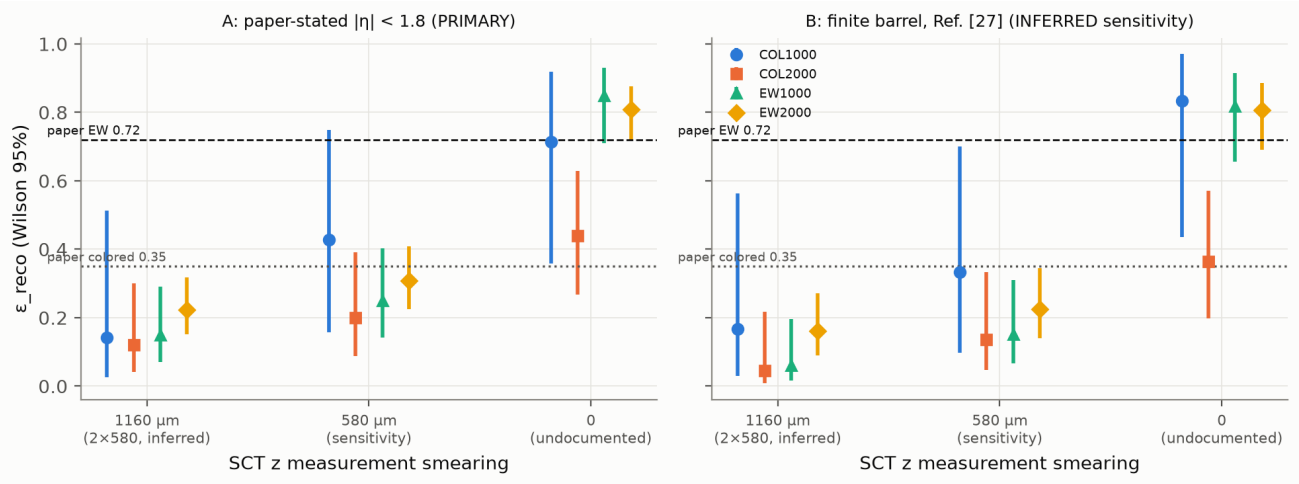
Reading

- Colored: trigger and fiducial agree. Reconstruction agrees for the 0.58 mm and no-smearing treatments, and is statistically inconsistent for 1.16 mm in the larger sample (3/25, p = 0.019).
- EW trigger: 17–18% against 10%. 36 one-at-a-time generator changes keep it at 13–19% (next figure); the same definition gives the paper's colored value, so an undocumented generator setting is the likely cause.
- EW fiducial: with the paper-stated |η| < 1.8 our value is 45–51% against 28% (strongly inconsistent). Only the finite barrel of [27] gives 28% (33–37%), but the paper does not say it used it. Unresolved.
- EW reconstruction: 15–31% with any SCT z smearing that has a source; 81–85% (borderline, p ≈ 0.07) only without SCT z smearing, which no source supports.

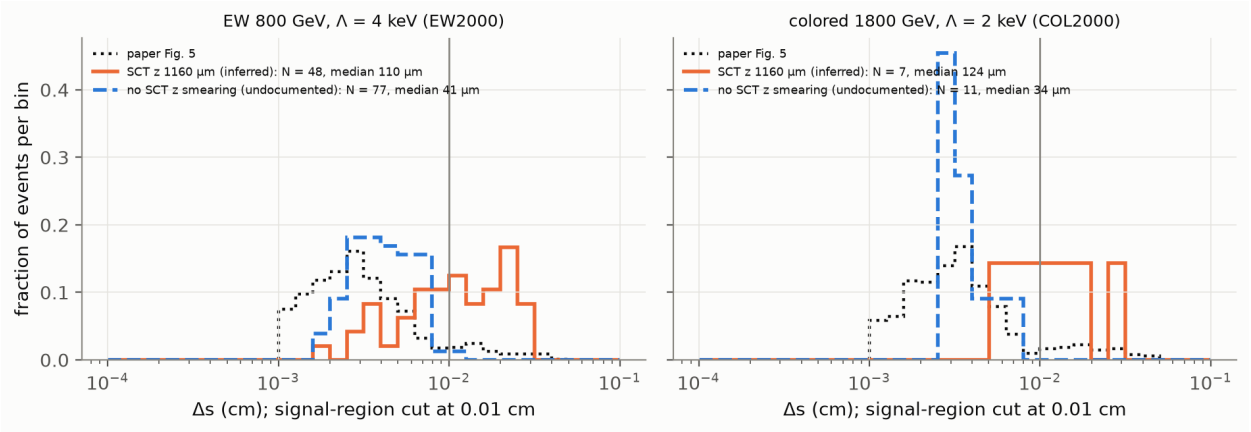


Left: EW vs colored trigger efficiency for every generator variation tried; star = paper. Right: all EW values.

5. What decides reconstruction: SCT resolution along the beam, and the beam spot



Reconstruction efficiency of the same events for each SCT z treatment. Left: paper-stated acceptance (main). Right: finite barrel (sensitivity).



Signal Δs before the last cut, paper-stated acceptance. With 1.16 mm smearing (solid) our events sit right of the paper's Fig. 5 (dotted); without it (dashed) they move close to it.

Reading

- Reference [27] gives the SCT accuracy as 17 μm across and 580 μm along the beam. The paper doubles “the intrinsic accuracy” without naming a coordinate, so 1.16 mm along the beam is our reading, not a stated number.
- With 1.16 mm the fitted plane is thick (signal Δs median 110–124 μm, cut at 100 μm). The paper's own Fig. 5 shows ~30 μm, which needs little or no SCT z smearing. The paper does not say what it ran.

The 45 mm beam spot. The paper applies “a uniform, gaussian smearing with width 45 mm to the z-coordinates of all the hits” and assumes the primary vertex “is identified correctly, and is located at the origin”. Three readings, counts as fiducial / final for SCT z 1.16 mm · 0.58 mm · none ($|\eta| < 1.8$):

Beam-spot reading (all inferred)	Colored 2000	EW 2000
One common shift per event, fit from the true vertex (used above)	25 / 3 · 5 · 11	94 / 21 · 29 · 76
One common shift per event, fit from the origin	25 / 0 · 0 · 0	94 / 3 · 3 · 4
Independent 45 mm shift of every hit	21 / 0 · 0 · 0	66 / 0 · 0 · 0

Per-hit smearing (45 mm ≈ 450× the Δs cut) leaves no reconstructable plane at all; a fit from the origin misses planes that pass through the displaced vertex. We use the common shift with the vertex fit because the paper says the vertex is identified correctly; this choice was not based on matching Table I.

6. Table I: paper vs this work (paper-stated acceptance)

	Quantity	Paper	Sample 1 (1000 gen.)	Sample 2 (2000 gen.)	Category (sample 2)
Colored	ε_trig	0.24	101/419 = 0.241 [0.203, 0.284] p = 0.95	207/873 = 0.237 [0.210, 0.266] p = 0.87	**Consistent**
	ε_fid (|η| < 1.8)	0.28	7/27 = 0.259 [0.132, 0.447] p = 1.00	25/68 = 0.368 [0.263, 0.486] p = 0.14	**Consistent**
	ε_reco, z 1.16 mm	0.35	1/7 = 0.143 [0.026, 0.513] p = 0.43	3/25 = 0.120 [0.042, 0.300] p = 0.019	**Statistically inconsistent**
	ε_reco, z 0.58 mm	0.35	3/7 = 0.429 [0.158, 0.750] p = 0.70	5/25 = 0.200 [0.089, 0.391] p = 0.14	**Consistent**
	ε_reco, no z smear	0.35	5/7 = 0.714 [0.359, 0.918] p = 0.056	11/25 = 0.440 [0.267, 0.629] p = 0.40	**Consistent**
EW	ε_trig	0.10	89/511 = 0.174 [0.144, 0.209] p = 2.8×10^{-7}	186/1045 = 0.178 [0.156, 0.202] p = 1.8×10^{-14}	**Strongly inconsistent**
	ε_fid (|η| < 1.8)	0.28	40/89 = 0.449 [0.350, 0.553] p = 8.2×10^{-4}	94/186 = 0.505 [0.434, 0.576] p = 1.3×10^{-10}	**Strongly inconsistent**
	ε_reco, z 1.16 mm	0.72	6/40 = 0.150 [0.071, 0.291] p = 9.1×10^{-14}	21/94 = 0.223 [0.151, 0.318] p = 2.4×10^{-23}	**Strongly inconsistent**
	ε_reco, z 0.58 mm	0.72	10/40 = 0.250 [0.142, 0.402] p = 9.4×10^{-10}	29/94 = 0.309 [0.224, 0.408] p = 1.5×10^{-16}	**Strongly inconsistent**
	ε_reco, no z smear	0.72	34/40 = 0.850 [0.709, 0.929] p = 0.078	76/94 = 0.809 [0.717, 0.875] p = 0.065	**Borderline / mild tension**

Wilson 95% CI in brackets; exact p vs the paper. The category uses the larger sample.

Bottom line

- Reproduced with paper-stated settings: colored trigger, colored fiducial, and the absence of background in the signal region.
- Not reproduced: EW trigger (17–18% vs 10%), EW fiducial with the stated |η| < 1.8 (45–51% vs 28%), and EW reconstruction with any SCT z smearing that has a source (15–31% vs 72%).
- Colored reconstruction depends on the SCT z treatment: 12–14% at 1.16 mm, 20–43% at 0.58 mm, 44–71% without smearing (paper 35%).
- A full reproduction is impossible from the published information: the acceptance actually used, the SCT resolution along the beam, the beam-spot treatment and the generator settings are not uniquely given, and the paper publishes no counts or errors.

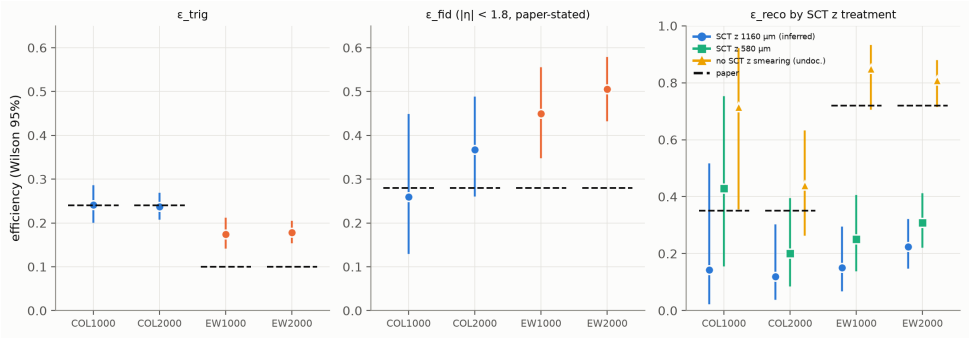


Table I comparison with the paper-stated acceptance; reconstruction shown for each SCT z treatment. Bars = Wilson 95%; dashed = paper.

Sensitivity study: finite barrel lengths of [27] (inferred, not the main result)

Sample	Fiducial	p vs 0.28	Final: z 1.16 mm · 0.58 mm · none
COL1000	6/27 = 0.22	0.67	1 · 2 · 5
COL2000	22/68 = 0.32	0.42	1 · 3 · 8
EW1000	33/89 = 0.37	0.059	2 · 5 · 27
EW2000	62/186 = 0.33	0.12	10 · 14 · 50

Appendix A. Where every setting comes from

Colour: green = stated in the paper or [27]; amber = inferred; red = undocumented. Counts: fiducial or final as noted, samples in the order COL1000, COL2000, EW1000, EW2000 where four are given.

Implementation choice	Exact source	Status	Resulting counts
Barrel only (pixel + SCT, 8 layers)	Paper Sec. III A: 'we only included the barrel of the pixel and SCT detectors in our simulations'	**PAPER-STATED**	frame for all counts
Acceptance: effective |eta| < 1.8, per hit, no finite-z boundary (PRIMARY)	Paper Sec. III A: '... which effectively restricts the fiducial range to |eta| <~ 1.8' (applying it as a per-hit cut is the implementation)	**PAPER-STATED**	fiducial COL1000 7/27, COL2000 25/68, EW1000 40/89, EW2000 94/186
Acceptance: finite barrel lengths (pixel |z|<400.5 mm, SCT |z|<749 mm, IBL |z|<330.15 mm)	Ref. [27] (arXiv:0901.0512 pp. 40-43) table 'Sensitive barrel 0 < |z| < 400.5 / 749'; IBL arXiv:1803.00844. No evidence the authors applied them	**INFERRED (sensitivity)**	fiducial COL1000 6/27, COL2000 22/68, EW1000 33/89, EW2000 62/186
Layer radii IBL 33.25, pixel 50.5/88.5/122.5, SCT 299/371/443/514 mm	Ref. [27] ranges '50.5<R<122.5', '299<R<514'; ATLAS JINST 3 S08003; IBL from arXiv:1803.00844 and paper Fig. 4 spacing	**REFERENCE-STATED (IBL INFERRED)**	all counts
Pixel hit = centre of hit pixel, 50x400 um (IBL 50x250 um)	Paper: 'the center of each pixel hit is used', 'pixel size rather than the resolution'; Ref. [27]: 'minimum pixel size on a sensor is 50x400 um2'	**PAPER-STATED + REFERENCE-STATED (IBL pitch INFERRED)**	all counts
Neighbouring-pixel merging	Paper: 'Hits in neightboring pixels ... are merged to a single hit' (adjacency rule INFERRED)	**PAPER-STATED**	all counts
SCT R-phi resolution 34 um	Paper: 'we conservatively assume a resolution of twice the intrinsic accuracy quoted in [27]'; Ref. [27]: '17 um (R-phi)'	**PAPER-STATED rule x REFERENCE-STATED value**	all counts
SCT z resolution 1160 um (= 2 x 580 um)	Same paper sentence (no coordinate named); Ref. [27]: 'intrinsic accuracies per module in the barrel are 17 um (R-phi) and 580 um (z)', footnote 'Arises from the 40 mrad stereo angle'. Original project choice	**INFERRED (paper rule applied to the reference z value; contradicted by paper Fig. 5)**	final COL1000 1/7, COL2000 3/25, EW1000 6/40, EW2000 21/94
SCT z resolution 580 um (1 x Ref. [27])	Ref. [27] value without the paper's doubling	**INFERRED (sensitivity; contradicts 'twice')**	final COL1000 3/7, COL2000 5/25, EW1000 10/40, EW2000 29/94
No SCT z measurement smearing	No source; closer to paper Fig. 5 Delta_s	**UNDOCUMENTED**	final COL1000 5/7, COL2000 11/25, EW1000 34/40, EW2000 76/94
45 mm longitudinal beam-spot smearing exists	Paper: 'a uniform, gaussian smearing with width 45 mm to the z-coordinates of all the hits'	**PAPER-STATED**	—
Beam spot = one common z shift per event, fit relative to the true vertex (PRIMARY reading)	Same sentence + 'We will assume that the primary vertex is identified correctly'	**INFERRED (wording ambiguous)**	SCT z 1160: COL2000 3/25, EW2000 21/94
Beam spot = one common z shift per event, fit relative to the origin	'... and is located at the origin' read literally	**INFERRED (alternative reading)**	SCT z 1160: COL2000 0/25, EW2000 3/94
Beam spot = independent 45 mm shift of every hit	'smearing ... to the z-coordinates of all the hits' read per hit	**INFERRED (alternative reading)**	fiducial COL2000 21/68, EW2000 66/186; final 0 and 0 for every SCT z (no plane found)
25 ns timing cut	Paper Sec. IV (location of the cut INFERRED)	**PAPER-STATED**	all counts
Delta_w = sqrt(lambda_2) (Eq. 11)	Paper Eq. 11; paper Fig. 4 band corresponds to 0.5 sqrt(lambda_2)	**PAPER-STATED (paper-internal inconsistency)**	no count changes

Implementation choice	Exact source	Status	Resulting counts
Trigger: pT(QQbar) > 200 GeV over matched events	Paper Sec. III B	**PAPER-STATED**	COL1000 101/419, COL2000 207/873, EW1000 89/511, EW2000 186/1045
Generator settings (matching scale, PDF, scales, tune)	not stated in the paper	**UNDOCUMENTED**	trigger as above; 36 variations give EW 0.13-0.19
Converted photon under acceptance A, PV fit origin	Paper Sec. III D (procedure); acceptance A as above	**as above**	final 0/9139 (SCT z 1160), 0/9139 (580), 30/9139 (no z smearing)

Appendix B. Paper-figure reproductions

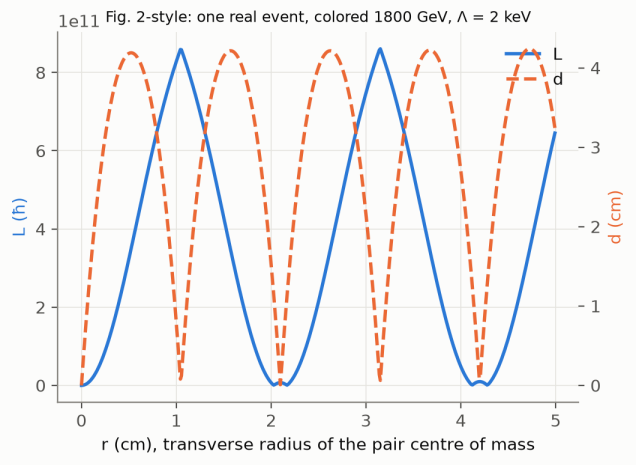


Fig. 2 style: angular momentum L and quirk separation d (colored, COL2000 event 107). L peaks near 9×10^{11} ħ, the paper's ~10^{12} ħ scale.

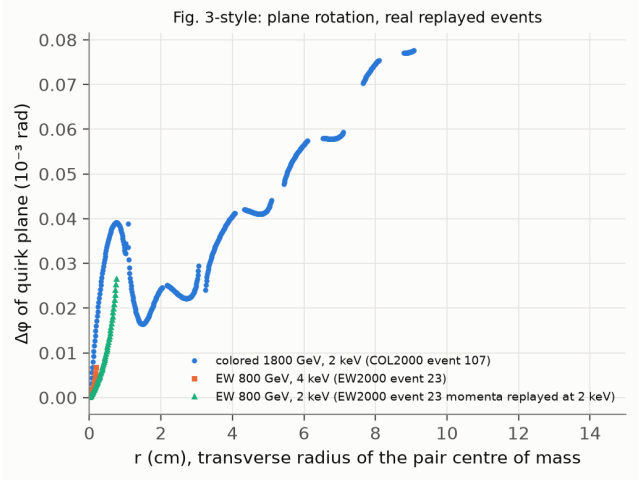


Fig. 3 style: rotation of the quirk plane stays below 10^{-4} rad, so the hits stay on a plane.

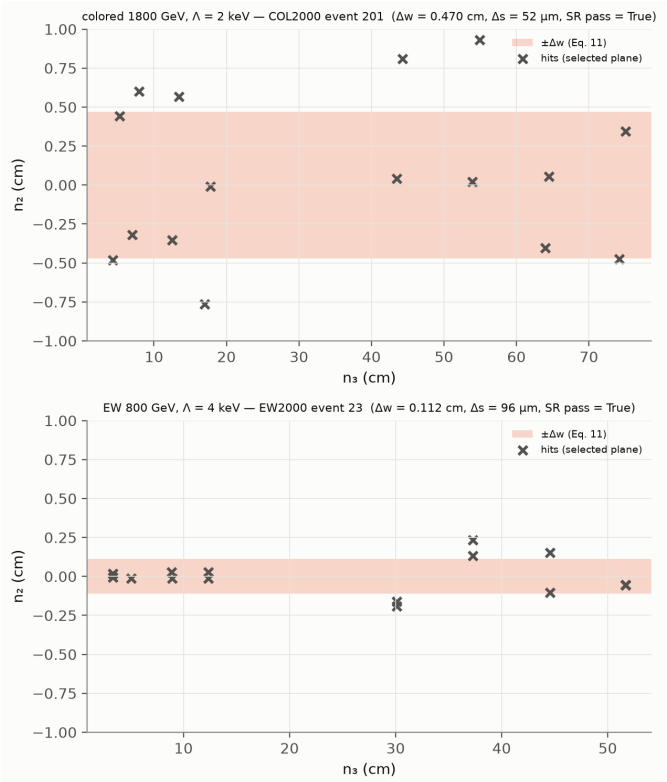


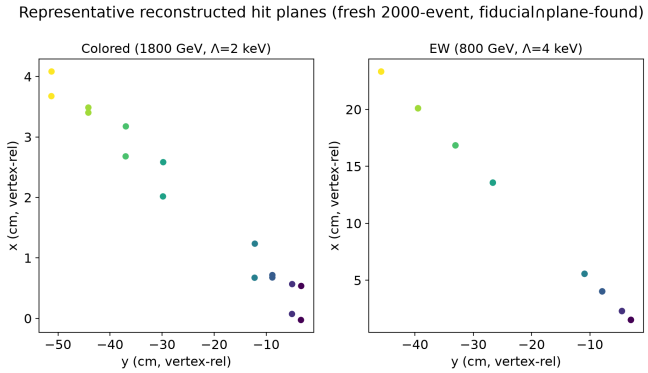
Fig. 4 style: one colored (COL2000 event 201) and one EW (EW2000 event 23) event, SCT z 1.16 mm. Both pass the signal region with either acceptance (Δs = 52 and 96 μm).

Appendix C. Check of the 30 September figure folder

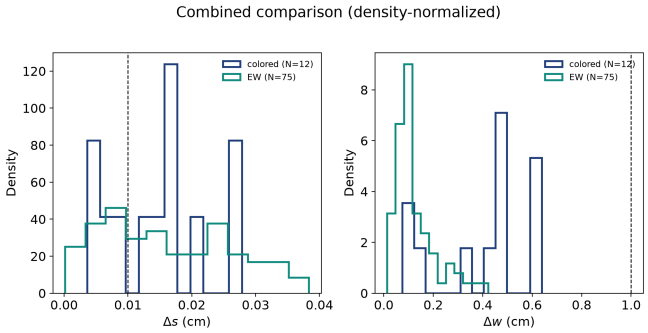
Folder `FINAL_FIGURES_2000_EVENT_2026-09-30`. Every number was recomputed from the per-event results. These figures use the paper-stated |η| < 1.8 acceptance with SCT z 1.16 mm, so their counts are exactly the main-report (v2) counts for that treatment.

Item	30 Sep figure (fid / plane / mult / final)	Recomputed now	Check
COL1000	7 / 5 / 3 / 1	7 / 5 / 3 / 1	**Match**
COL2000	25 / 12 / 7 / 3	25 / 12 / 7 / 3	**Match**
EW1000	40 / 32 / 16 / 6	40 / 32 / 16 / 6	**Match**
EW2000	94 / 75 / 48 / 21	94 / 75 / 48 / 21	**Match**
Fig. 5 colored values	12 events	12/12 identical	**Match**
Fig. 5 EW values	75 events	57/75 identical; 17 differ by ≤ 30 μm and 1 has no plane on this machine (floating-point replay); 0 pass/fail changes at the Δs cut	**Match**
Fig. 4 events (65, 182)	Δs, Δw	identical	**Match**

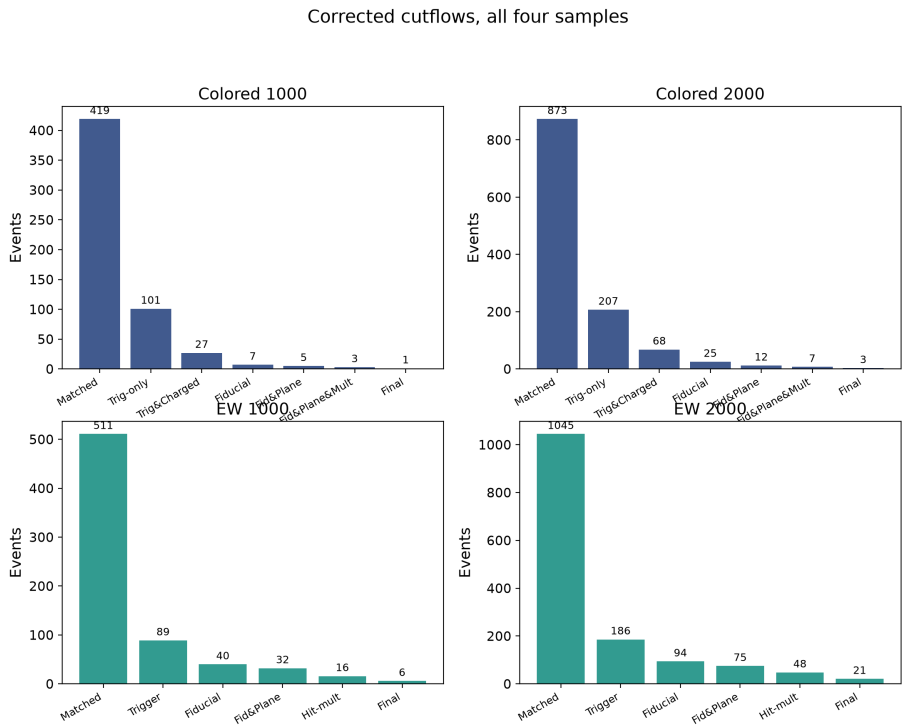
The 30 Sep Fig. 2 is not reused: it computed L with lab-frame momenta and reached ~5×10^{13} ħ. Computed in the quirk pair's rest frame (Appendix B) it is ~9×10^{11} ħ, in line with the paper.



30 Sep Fig. 4 (|η| < 1.8, SCT z 1.16 mm): hit positions of colored event 65 and EW event 182.



30 Sep Fig. 5 (|η| < 1.8, SCT z 1.16 mm): Δs and Δw of the 12 colored and 75 EW events with a plane.



30 Sep cutflows of all four samples (|η| < 1.8, SCT z 1.16 mm); counts verified in the table above.